Ahmedabad University

PSY310

Reconstructive Memory Experiment

Adaptation of Loftus & Palmer's Study (1974)

Solo Project

Submitted to faculty: Prof. Nithin George

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Git-Hub Link:

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Introduction

Overview

Memory is the cognitive process of the encoding, storage, and retrieval of information. From the 1800s to the early 1900s, this process was believed to work as a “storage system” of information and recording, and forgetting occurred due to simple decay over time.[1] However, after the schema theory was introduced in 1932, doubts were raised upon the idea that memory is a flawless process of the recording of information. The schema theory defines “schemas” as internal representations which help us organise and interpret information through the lens of our beliefs and past experiences.[2] These schemas are also created based on post-event experiences and irrelevant sensory information which cause us to fill in the gaps of lost pieces of information with false information, distorting our memory of the event.[3]

This led to the proposal of the theory of reconstructive memory. It states that the retrieval of memory, instead of a passive process of the recollection and decay of information, is an active process involving the reconstruction and restructuring of information based on schemas.[4]

Loftus & Palmer (1974)[5] conducted a study on the effect of leading questions on eyewitness testimony, which served as a foundation for the theory of reconstructive memory. They split participants into 5 groups and showed each of them 7 films of motor accidents. After each film, a questionnaire was given to each participant, which contained mostly distractor questions, but had one critical question: “About how fast were the cars going when they *hit* each other?” In each condition, the verb ‘hit’ was replaced with a verb of higher or lower intensity, such as ‘contacted’ or ‘smashed.’ They hypothesized that participants would estimate a higher speed when presented with a verb of higher intensity in the critical question. Results matched their hypothesis, showing a significantly higher speed estimated by participants in the high intensity condition than the low intensity condition.

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[1] Ebbinghaus, 1885
[2] Carbon & Albrecht, 2012
[3] Bartlett, 1932
[4] Loftus, 1981
[5] Loftus & Palmer, 1974

Effect of extraneous variables on memory recall

Eyewitness memory is reconstructive memory and very sensitive to suggestive influence, that is, extraneous variables, like color, can influence attention, arousal, and source monitoring in a manner that distorts recollection and confidence especially when vivid and high-intensity colors co-occur with leading language.[6] Colour-memory research, reviewed, suggests that warm, saturated colour (e.g. reds) is predictable in increasing attention and the perceived vividness of the contents encoded, and therefore increases the weight of perceptual features during subsequent reconstruction, which can bias the response given under suggestive questioning to what feels salient as opposed to what is accurate.[7] In modern theories of false memory, encoding or retrieval arousal or perceptual richness will facilitate gist-based retrieval and misidentification of sensory information (e.g., colour and location), increasing the probability that the proposed or similar features will be added to an existing, though inaccurate, memory trace, particularly when cues are conceptually elaborated or placed in evocative contexts.[8] Furthermore, particular color effects like the primacy of red to indicate significance can differentially enhance confidence and binding of color characteristics to objects, a process that can enhance memory distortion when high intensity colors are matched with high intensity verbs, thereby aligning attentional capture, arousal and suggestibility in a manner that biases eyewitness reports in a systematic manner.

Hypothesis

The intensity of the verb used in the critical question and the intensity of the background colour will have a significant impact on the speed of the vehicle estimated by the participant and recalling whether they saw broken glass. Participants who are presented with the high intensity verb and background colour will estimate a higher speed than the participants who are presented with the low intensity verb and background colour.

On the other hand, the intensity of the verb and background colour may not affect participant’s speed estimates, which would result in similar speed estimates from participants presented with either condition.

[6] Dzulkifli & Mustafar, 2013

[7] Kuhbandner et al., 2015

[8] Zaragoza et al., 2011

Method

Participants

A total of 12 Ahmedabad University undergraduate students (N=12; 6 women and 6 men; mean age = 19.1667 years; standard deviation of age = 1.4035 years) participated in this experiment. The equal number of male and female participants in each condition helps improve the generalizability of the findings. Each participant was informed of the experiment's procedure and voluntarily participated. After the experiment was completed, each participant was debriefed about the true aim of the study. No reward or payment was given in exchange for their participation.

Apparatus & Stimuli

Two videos of car accidents were downloaded from YouTube (See Appendix A). Both videos are 9 seconds long. The experiment was created on a Lenovo ThinkPad with a resolution of 1536 x 960 pixels using the PsychoPy (v2025.1.1) software.

Design

The experiment is a between-subjects design with two independent variables. The first independent variable is the intensity of the verb in the critical question (high, low) and the second independent variable is the intensity of the background colour (high, low). The high intensity verb used in the critical question is "demolished" and the low intensity verb is "contacted;" the high intensity background colour is crimson (hex code: #dc143c) and the low intensity background colour is lavender (hex code: #e6e6fa). There are two dependent variables; firstly, the speed (in km/h) of the vehicle in the car crash estimated by the participant after viewing the first video, and secondly, whether the participant remembered seeing glass break after viewing the second video.

Procedure

The experiment began with a screen which instructed participants that they will be shown two videos of car accidents, followed by a few questions. After that, the first video began and played for 9 seconds. Then, a distractor question was asked: "What was the colour of the car in the video?" Next, the critical question was asked: "About how fast was the car going when it *demolished* the other car?" in which the intensity of the verb "demolished" is high. Thirdly, another distractor question was asked: "How many total cars did you see in the video?" Following a transition screen, the second video of a car crash was played for 9 seconds. Then, two distractor questions were asked: "Was it the white car's fault?" and "How was the weather like in the video?" Next, the critical question was presented to the participant: "Did you see any broken glass?" where the participant was asked to click the 'Y' key if they remembered seeing broken glass, or the 'N' key if they did not remember seeing any broken glass.

To conclude the experiment, the debriefing screen thanked the participants for their participation, informed them about the hypotheses, the true aim of the experiment and that they can ask to view their results if they wish to. The participants’ responses were recorded in an Excel format by the PsychoPy software.

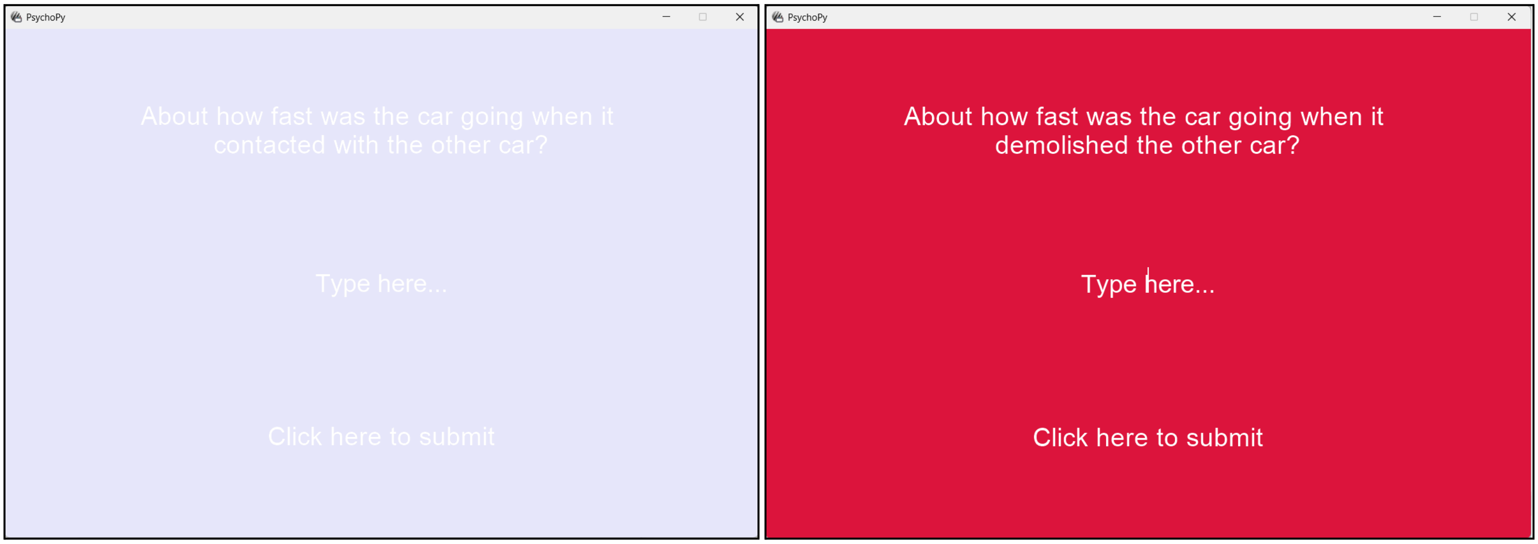

Figure 1 - First critical question screens in low intensity and high intensity conditions

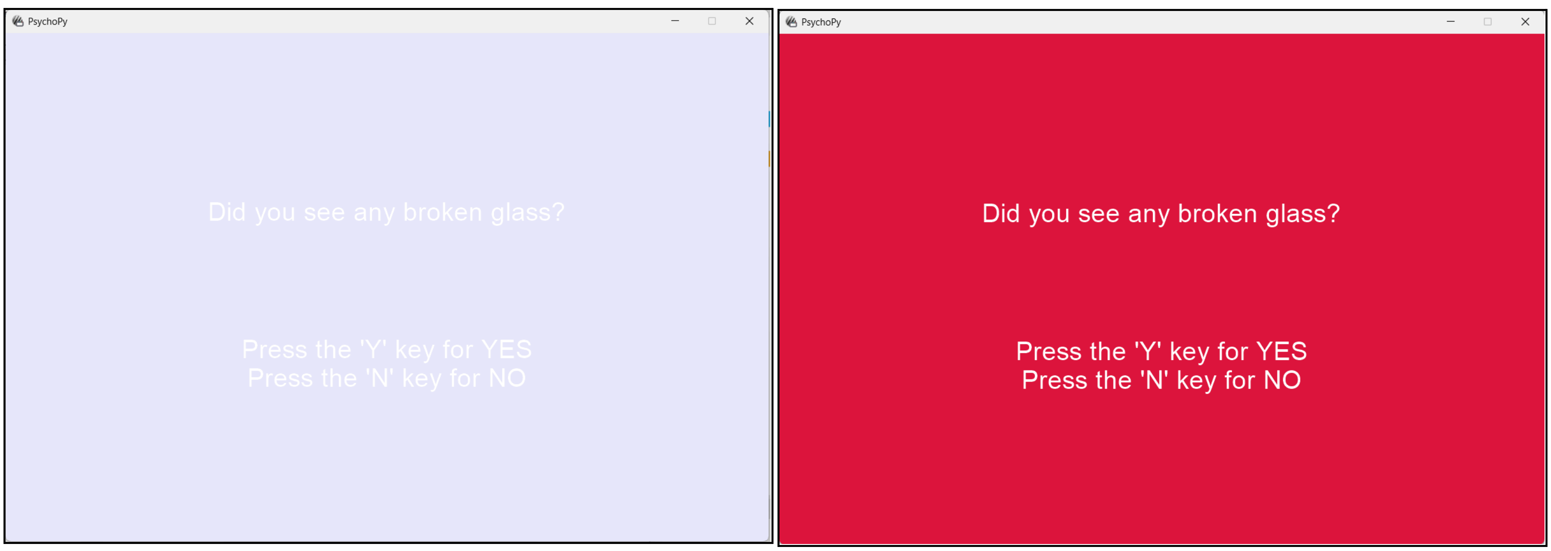

Figure 2 - Second critical question screens in lower intensity and higher intensity conditions

Analysis & Results

Descriptive Statistics

Condition	Mean speed estimate (in km/h)	Median	Standard Deviation
Contacted	47.5	45	10.83974169
Demolished	65	60	16.43167673

Figure 3 - Table of Descriptive Statistics

Results show that the intensity of the verb in the critical question and the background colour have an impact on the speed estimated by the participants. The mean speed estimates between the participants of both conditions is 17.5 km/h, and a 15 km/h difference between the medians.

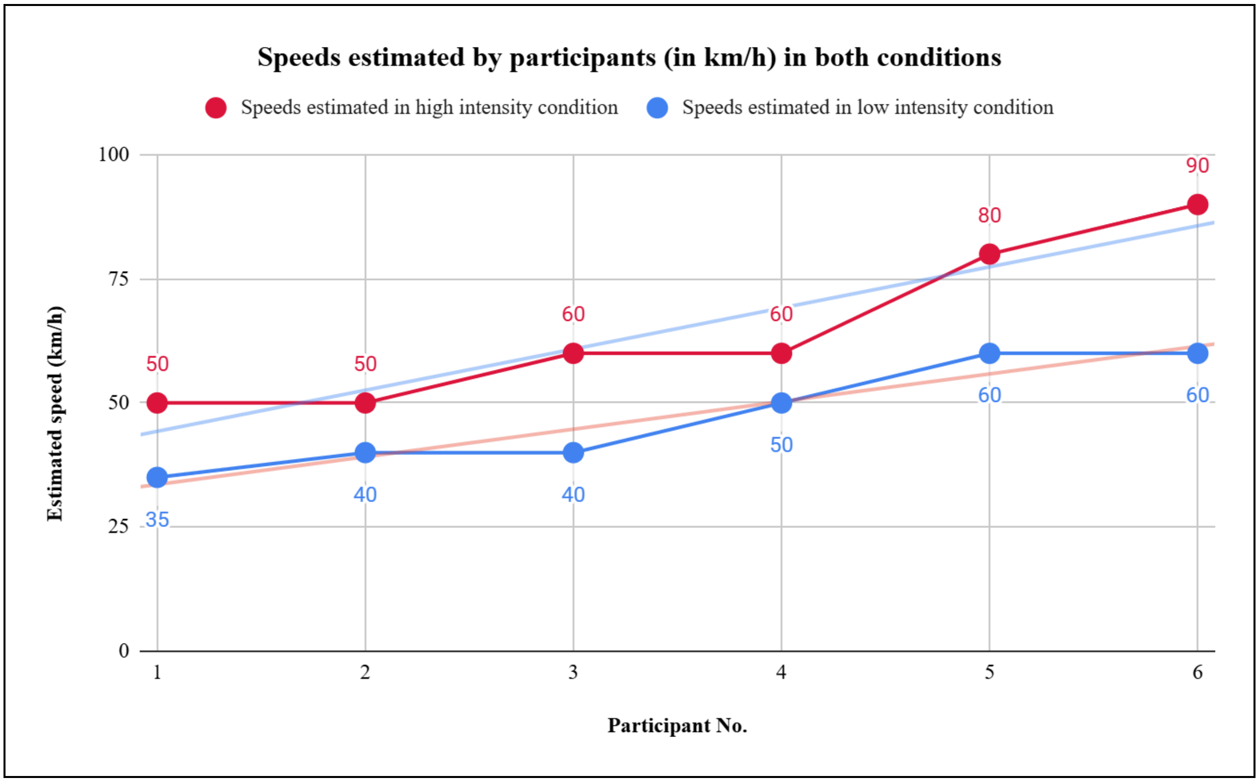

Figure 4 - Scatter plot of speeds estimated by each participant

This scatter plot shows that two participants from the low intensity condition estimated an equal or higher speed than four participants of the high intensity condition. Thus, this highlights that the hypotheses may not be of concrete strength, or the study design can be improved.

Results also showed that only one out of the 12 participants answered “Yes” to the question “Did you see any broken glass?” This indicates that there was little to negligible observable effect of high or low intensity verbs and background colour on distorting the memory of the participants into leading them to think that they saw broken glass, despite there being two distractor questions between the participant viewing the video and facing the critical question. This may be due to the video very evidently showing no broken glass, or the study design could be further improved.

Inferential Statistics

A one-tailed two-sample t-test was performed to examine the effect of the intensity of the verb in the critical question and the intensity of the background colour on the speed of the vehicle estimated by the participant.

H_0: $\mu_{\text{contacted, lavender}} \geqq \mu_{\text{demolished, crimson}}$

H_A: $\mu_{\text{contacted, lavender}} < \mu_{\text{demolished, crimson}}$

The one-tailed two-sample t-test indicates statistically significant effects at the 0.05 significance level of the intensity of the verbs and the background colour on the speed of the vehicle estimated by the participants. A p-value of 0.0272 was attained as speed estimated by participants in low intensity conditions (mean = 47.5, standard deviation = 10.8397) were significantly lower than participants in high intensity conditions (mean = 65, standard deviation = 16.4317).

Participant No.	Speeds estimated in high intensity condition	Speeds estimated in low intensity condition
1	50	35
2	50	40
3	60	40
4	60	50
5	80	60
6	90	60
p-value	0.0272330627	

Figure 5 - One-tailed two-sample t-test performed using the TTEST command on Google Sheets

Discussion

The experiment helps us to understand more about reconstructive memory by supporting the idea that the process of recalling memories is not like a “storage system”, but is a constructive process, which depends on contextual and linguistic factors. The findings revealed that the individuals who were exposed to high intensity verb and color of the background made a significantly higher estimate of the speed of the vehicle compared to those who were exposed to the low intensity condition. This can be correlated with the schema theory and the results of Loftus and Palmer’s 1974 study on eyewitness testimony who prove that memory is partially formed in terms of expectations and emotional interest instead of flawless recollection. Background color as an extraneous variable is an addition to the traditional research which has shown the interaction of visual arousal with linguistic suggestibility and that the color intensity can create more distortion during retrieval. However, this experiment was unable to prove a relationship solely between colour intensity and memory recall, as the factor of the intensity of the verb could not be differentiated from the intensity of the colour. Also, as there was no significant difference in whether the participants remembered seeing glass breaking, further research is still required to determine the true impact of colour on memory recall.

One of the strengths of this study is that it combines both perceptual (color) and linguistic (verb intensity) variables that can be understood more comprehensively regarding how a combination of several cues works together to affect memory. The study has an experimental control, including equal gender proportion and the length of stimuli being the same, which increases internal validity. However, there are a number of weaknesses that do not allow the generalization of the findings. The sample size of 12 participants is very small, which severely limits the statistical power and exposes the research to random variation. The two color conditions are used and the number of videos is rather small, limiting ecological validity and the possibility to draw accurate conclusions about any strong effect in different situations. Also, the false recall of broken glass is low is an indication of the ceiling effect or weak manipulations to cause distortion of memory other than that of speed estimation. Future research can enhance them by adding more samples, varying the stimuli, and physiological measures of arousal to more effectively reveal the interaction of color and language in biasing memory retrieval.

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Appendix A

GitHub Link:
https://github.com/aryanm4/PSY310_LabInPsychology/tree/Reconstructive-Memory-(Solo-Project)

Video 1:
https://github.com/aryanm4/PSY310_LabInPsychology/blob/Reconstructive-Memory-(Solo-Project)/Video%201.mp4

Video 2:
https://github.com/aryanm4/PSY310_LabInPsychology/blob/Reconstructive-Memory-(Solo-Project)/Video%202.mp4

Appendix B - Ethical Clearance Application Form

Details of Project

Sr. No.	Title	Particulars
1	Title of research project	Reconstructive Memory Experiment Adaptation of Loftus & Palmer's Study (1974)
2	Anticipated duration of the project/trial	7 days
3	List and details of location(s) where project will be conducted [Include all the collaboration with institute/hospital/practitioner]	The experiments will be performed inside the psychology lab at the School of Arts and Sciences, Ahmedabad University
4	Source of funding & financial allocation for the project / trial	This research is not funded by any agency.
5	Will the research proposal be submitted to any ministry screening committee for international collaboration?	No
6	Brief summary of research proposal	This study examines the impact of leading questions and extraneous variables on the memory recall of eyewitnesses of a car accident. The intensity of verbs used and background colour is hypothesized to impact the speed of a vehicle estimated by the participant, and whether they remember

		viewing broken glass.
7	Research objectives	1) To understand the impact of extraneous variables on memory recall. 2) To observe how the theory of reconstructive memory plays a role in eyewitness testimony.
8	Methodology and Technical approach of the proposal	The study follows a quantitative and experimental methodology using a between-subject design. The variables of interest are manipulated in a controlled environment. Participants are shown two videos of car accidents and asked a few distractor questions and two critical questions. In the first condition, the verb used in the critical question will be of high intensity: "demolished" and low intensity in the second condition: "contacted." The high intensity background colour is crimson, and the low intensity colour used is lavender. Stimuli will be presented using PsychoPy on a Windows PC. The speed of the vehicle in the accident estimated by the participant and whether they remember viewing broken glass in the video will be recorded by the software in an Excel format.
9	Usefulness of the project / trial	The proposed experiment helps us understand how extraneous variables and leading questions can distort one's memory of an event, and how it plays a role in eyewitness testimonies.

Details of Investigator(s)

Sr. No.	Title	Particulars
1	Name of the Principal Investigator (PI) with Qualification, Designation, School, Email Address and, Phone Contact	Aryan Malkani, BA (Hons) Student, School of Arts and Sciences, Ahmedabad University aryan.m4@ahduni.edu.in +91 77009 70497

2	Name of the Co-Investigator(s) Co-PI) with Qualification, Designation, School, Email Address and, Phone Contact	NA
3	Name of Supervisor (In case of student involvement) Qualification, Designation, School, Email Address and, Phone Contact	Professor Nithin George, PhD Assistant Professor School of Arts and Sciences, Ahmedabad University nithin.george@ahduni.edu.in 8488865331
4	Name(s) of student involved with programme name and semester	Aryan Malkani BA (Hons) Student, School of Arts and Sciences, Semester 3
5	Name(s) and status of other investigators including students and professional collaborators	NA

Details of ethical consideration for research/project

Sr. No.	Title	Particulars
1	Is approval by_other Ethics Committee required? If 'Yes' mention the details.	Yes
2	Type of review requested	Exempted Review/ **Expedited Review**/ Full Committee Review
3	Justify why a human study is needed to answer the research questions.	This experiment aims to understand memory recall and distortion through extraneous variables, which is only possible with human participants.
4	Participant recruitment process and eligibility criteria	Participants will be recruited through volunteerism. Eligible participants will be between the ages of 18 and 25 and have normal or corrected-to-normal vision.
5	Number of participants required with necessary justification.	As per guidelines provided from the course instructor, the necessary number of participants is 10-12.
6	Justify inclusion and exclusion of vulnerable population	Individuals with visual deficits are excluded because the experimental task is

		screen-based and requires normal visual acuity to perform. Individuals with a below average proficiency in understanding English commands are excluded as the experiment requires understanding questions in English.
7	Procedure for seeking and obtaining informed consent with a sample of the patient/participant information sheet and informed consent forms in English and local languages. [AV recording if needed] Informed consent for stored samples	Verbal consent is taken.
8	Plan for statistical analysis of the study	The data obtained from the experiment (speeds estimated by participants in km/h) will be analyzed using a one-tailed two-sample t-test. Statistical Analysis will be performed using Microsoft Excel.
7	Explain the plans to maintain confidentiality of records / data of the study participants	All participant data will be number-coded, and the file will not be tagged with the participant's name. The data will be stored in a password-protected location. The participant’s name or any other sensitive information will not be made public.
8	Explain all anticipated risk (adverse events, discomfort) or injury that may be caused to the participant	There is no risk or benefit for the participant.
9	Efforts taken to minimize the risk or injury	The study involves tasks that are not likely to cause injury. Participants will be explicitly told that they can walk out of the experiment at any time.
10	Whether 'wage compensation' for the research subjects will be provided? [Compensation/ Reimbursement of incidental expenses and management of research related injury/illness]	No compensation is provided.
11	Expected 'benefits' to volunteer / community	No benefit to the volunteer.
12	Account of storage and maintenance of all the human material or data obtained from the study	The data will be stored in a password-protected private server.
13	Explain the plans of publication of results (positive or negative) while maintaining confidentiality of personal information/identity	Results will not be published.

14	Specific ethical issues, as identified by the investigating team	None.
15	List of documents enclosed for ethical review	None.
16	Disclose Conflict of Interest, if any.	None.